

Online Appendix to “Only Checking Where the Model Disagrees: Discrepant-Resolution Bias in Benchmark Relabelling”

This online appendix describes the data processing and the literature-coding protocol, and gives two tables and a figure that support Section 7 of the paper. Code and data are at <https://github.com/unluckyjet/discrepant-resolution>.

1 Data processing (MMLU-Redux and HELM)

MMLU-Redux 2.0 contains 5,700 items. The analysis set has 5,421 items: the 5,321 items annotated as correct and the 100 items annotated with a wrong ground truth and a parseable corrected answer. We exclude the following items:

- 131 with bad question clarity;
- 38 with multiple correct answers;
- 36 with no correct answer;
- 32 marked “expert”;
- 25 with bad option clarity;
- 5 marked as wrong ground truth whose suggested answer could not be parsed;
- 1 marked as wrong ground truth whose suggested answer equals the MMLU label;
- 10 that could not be matched to a HELM instance;
- 1 whose MMLU-Redux copy of the label differs from HELM’s.

The suggested answer appears in several formats (an index, a float, a letter); a parser handles all of them.

We downloaded HELM v1.3.0 per-instance predictions for all 37 models and 57 subjects (14,042 instances per model; method `multiple_choice_joint`). A prediction is parsed as an option letter only if the output begins with a bare letter A–D, optionally in parentheses and not followed by another letter. Every other output is invalid and counts as wrong under every label. Our parsed correctness agrees with HELM’s exact-match scores on all but 29 of the $37 \times 14,042$ instance predictions (at most 3 per model). The disagreements are items with duplicated option texts, which HELM scores by text.

Invalid outputs in the analysis set range from zero (for 22 models) to 994 (Mistral Large). In Scenario B an invalid labeler output becomes a deliberately wrong label. All scripts, the derived item-level table and the results files are released.

2 Literature coding protocol

Candidate studies were collected from Semantic Scholar searches for benchmark relabelling, label-error audits and benchmark cleaning, and from the reference lists of the studies found. For each study we read the methods section in full text. We then recorded the following:

- the adjudication design, in the categories of Section 6 of the paper;
- the flagging model or algorithm;
- whether a flagging model was also evaluated on the corrected labels;
- whether accuracies or rankings were re-reported;
- whether the paper acknowledges that unflagged items may be mislabelled.

Each code is supported by a verbatim quote with its page. A script checks every quote against the extracted full text, and all quotes matched. Confidence was recorded as high or medium; no study was coded without a supporting quote. Studies whose phases follow different designs are coded OTHER. The coding table, with quotes, is released as `data/lit/design_coding.csv`.

3 Additional tables

Flagger	True acc. (%)	Adjudicated	Hidden $ J $	Inflation (pp)	True rank	DR rank	Flagger-favouring
Claude 3 Opus	86.13	802	17	0.31	1	1	0
GPT-4o	85.78	819	19	0.35	2	2	0
GPT-4 (0613)	84.63	879	17	0.31	3	3	0
Gemini 1.5 Pro	82.92	973	20	0.37	4	4	0
GPT-4 Turbo	81.55	1045	16	0.30	5	5	0
Llama 3 70B	80.98	1075	19	0.35	6	6	0
PaLM 2 Unicorn	80.39	1109	19	0.35	7	7	0
Palmyra X v3	80.37	1110	19	0.35	8	8	0
Mixtral 8x22B	79.60	1147	22	0.41	9	9	0
Gemini 1.5 Flash	79.54	1148	20	0.37	10	9	1
Qwen1.5 72B	79.04	1182	16	0.30	11	11	0
Claude 3 Sonnet	77.81	1245	22	0.41	12	12	0
Yi 34B	77.64	1255	18	0.33	13	13	0
Qwen1.5 32B	76.00	1342	20	0.37	14	14	0
Claude 3 Haiku	75.74	1359	18	0.33	15	15	0
DBRX Instruct	75.61	1364	17	0.31	16	15	1
Claude 2.1	75.36	1383	19	0.35	17	16	1
DeepSeek 67B	73.88	1456	19	0.35	18	18	0
Mixtral 8x7B	73.53	1486	14	0.26	19	19	0
Gemini 1.0 Pro	71.87	1583	8	0.15	20	20	0
Llama 2 70B	71.15	1610	14	0.26	21	21	0
Mistral Large	70.98	1609	13	0.24	22	22	0
PaLM 2 Bison	70.93	1624	14	0.26	23	22	1
Mistral Small	70.93	1622	11	0.20	23	22	1
Qwen1.5 14B	70.84	1635	13	0.24	25	22	2
Claude Instant 1.2	70.65	1637	15	0.28	26	25	1
GPT-3.5 Turbo	70.39	1646	17	0.31	27	27	0
Arctic Instruct	69.06	1719	19	0.35	28	28	0
Llama 3 8B	68.86	1735	15	0.28	29	29	0
Gemma 7B	67.42	1807	17	0.31	30	30	0
Yi 6B	65.49	1910	16	0.30	31	31	0
Qwen1.5 7B	64.91	1952	11	0.20	32	32	0
Phi-2	59.31	2243	17	0.31	33	33	0
Mistral 7B	57.94	2314	17	0.31	34	34	0
Llama 2 13B	56.71	2379	17	0.31	35	35	0
Llama 2 7B	46.93	2918	13	0.24	36	36	0
OLMo 7B	29.42	3824	30	0.55	37	37	0

Table OA.1: Scenario A (original MMLU labels): DR audits for every single-model flagger, ordered by true accuracy.

Sc.	Model	True	$\varepsilon = 0$	$\varepsilon = 0.01$	$\varepsilon = 0.05$	$\varepsilon = 0.1$	$\varepsilon = 0.2$
A	Claude 3 Opus	86.1	[86.3, 86.3]	[85.5, 87.2]	[82.1, 90.6]	[77.8, 91.8]	[69.4, 91.8]
A	GPT-4o (F)	85.8	[86.1, 86.1]	[85.3, 86.1]	[81.9, 86.1]	[77.6, 86.1]	[69.1, 86.1]
A	GPT-4 (0613)	84.6	[84.9, 84.9]	[84.1, 85.8]	[80.7, 89.2]	[76.4, 90.4]	[67.9, 90.4]
A	Gemini 1.5 Pro	82.9	[83.1, 83.1]	[82.3, 84.0]	[78.9, 87.3]	[74.6, 90.9]	[66.1, 90.9]
A	GPT-4 Turbo	81.6	[81.8, 81.8]	[81.0, 82.7]	[77.6, 86.1]	[73.3, 86.4]	[64.8, 86.4]
A	Llama 3 70B	81.0	[81.2, 81.2]	[80.3, 82.0]	[76.9, 85.4]	[72.7, 89.7]	[64.2, 90.2]
B	Claude 3 Opus	86.1	[85.8, 85.8]	[85.0, 86.6]	[81.9, 89.7]	[78.0, 93.5]	[70.1, 93.5]
B	GPT-4o	85.8	[85.7, 85.7]	[85.0, 86.5]	[81.8, 89.7]	[77.9, 93.2]	[70.1, 93.2]
B	GPT-4 (0613)	84.6	[85.0, 85.0]	[84.2, 85.8]	[81.1, 88.9]	[77.1, 92.8]	[69.3, 92.8]
B	Gemini 1.5 Pro	82.9	[83.3, 83.3]	[82.5, 84.1]	[79.4, 87.2]	[75.4, 91.1]	[67.6, 91.5]
B	GPT-4 Turbo	81.6	[82.1, 82.1]	[81.3, 82.8]	[78.1, 86.0]	[74.2, 88.0]	[66.4, 88.0]
B	Llama 3 70B (F)	81.0	[90.7, 90.7]	[89.9, 90.7]	[86.8, 90.7]	[82.9, 90.7]	[75.0, 90.7]

Table OA.2: Sharp identified intervals (accuracy, %) for the six most accurate models and the flagger (F) under assumed bounds ε on the concordant label-error rate (Theorem 6 of the paper). Flaggers: GPT-4o in Scenario A, Llama 3 70B in Scenario B (Mixtral 8x7B labels).

4 The rate in m

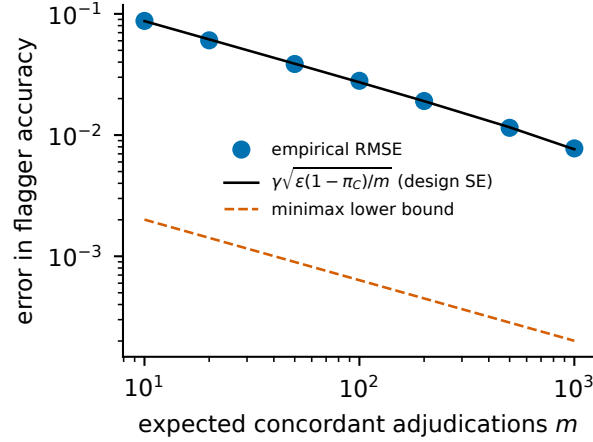


Figure OA.1: Error of the two-phase estimate of the flagger’s accuracy (Llama 3 70B, Scenario B with Mixtral 8x7B labels) against the expected number m of concordant adjudications, with $\pi_\Delta = 1$, over 2,000 replicates. The solid line is the design standard error (equation (3) of the paper); the dashed line is the minimax lower bound of Theorem 9 of the paper, evaluated at the observed $\gamma = 0.784$ and $\varepsilon = 0.124$.